

- Q.25 Write a short note on Blue tooth.
- Q.26 Write steps to open & disassemble Android smartphone.
- Q.27 Write short on GPRS.
- Q.28 Write the features of FDMA & TDMA systems.
- Q.29 Explain DS-CDMA technique.
- Q.30 Mention the name of tools used in open & disassemble mobile phone.
- Q.31 Write the feature of UTMS.
- Q.32 Explain the different databases (HLR, VLR & AuC) used in network and switching subsystem of GSM mobile.
- Q.33 Write short note on EDGE.
- Q.34 Explain the various interfaces used in GSM.
- Q.35 Explain the narrow band & wideband systems.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Compare the different generations of mobile communication system in tabular form.
- Q.37 Define CDMA & give comparison between GSM & CDMA system.
- Q.38 Define wireless communication. Write the advantages & disadvantages of wireless communication.

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#### 6th Sem / Eltx

#### Subject:- Wireless and Mobile Communication / Digital & Data Communication

Time : 3Hrs. M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 UHF are in the range of  
a) 30MHz to 300 MHz b) 30 KHz to 300 Khz  
c) 300MHz to 300MHz d) 300MHz to 3GHz
- Q.2 The total Channel capacity for cellular telephone area is given by  
a)  $C = mGN$  b)  $C = m(GN)^{1/2}$   
c)  $C = m(GN)^3$  d)  $C = m(GN)^{1/3}$
- Q.3 PCS is a \_\_\_\_\_ Generation wireless system  
a) 1G b) 2G  
c) 3G d) 4G
- Q.4 IMTS Stands for  
a) Improved Mobile Telecommunication Services  
b) Improved Mobile Telephone Services  
c) Integrated Mobile Telecommunication Services  
d) Intelligent Mobile Telecommunication Services

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Q.5 \_\_\_\_\_ multiple access technique is used in AMPS.

- a) TDMA                      b) SDMA
- c) CDMA                     d) FDMA

Q.6 If PFO is faulty in mobile cell then

- a) there will be no network in mobile phone
- b) displays call end
- c) software problems in the mobile & set will get dead
- d) All of the above

Q.7 The radius of Pico cell is

- a) Less than 400ft
- b) Between 400ft to 2000ft
- c) Above 2000ft
- d) All the above

Q.8 \_\_\_\_\_ is the process of subdividing a congested cell into smaller cell.

- a) Cell sectoring              b) Cell splitting
- c) Frequency reuse          d) none of the above

Q.9 FDMA is the division of

- a) Time                        b) Frequency
- c) Phase                      d) Amplitude

Q.10 \_\_\_\_\_ modulation is used in GSM.

- a) QPSK                        b) FSK
- c) GMSK                      d) ASK

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Data rate in 1G cellular system is \_\_\_\_\_.

Q.12 Define hand off.

Q.13 Expand PSTN

Q.14 Expand two application of Wi-Fi.

Q.15 GSM stands for \_\_\_\_\_

Q.16 Name two devices in which Bluetooth technology is used.

Q.17 Expand GPS.

Q.18 Expand MSC.

Q.19 What happens if RAM of a mobile phone is found faulty.

Q.20 FM broadcasting uses \_\_\_\_\_ frequency range.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the various sections of a mobile phone.

Q.22 Explain Multipath effect.

Q.23 What do you mean by center excited cell & edge excited cell.

Q.24 Explain HPSA with its advantages & disadvantages.